

Plots and tables

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```
# SCRIPT FOR DATA NOTE GRAPHS
# libraries ----
library(openxlsx)
library(tidyverse)

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.2.0      v readr      2.2.0
## v forcats    1.0.1      v stringr    1.6.0
## v ggplot2     4.0.2      v tibble     3.3.1
## v lubridate  1.9.5      v tidyr      1.3.2
## v purrr      1.2.1
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

library(patchwork)

# ggplot theme ----

my_theme <- function(...){
  theme_minimal() +
    theme(axis.title.y = element_text(size = 5),
          axis.title.x = element_text(size = 15),
          axis.text.y = element_text(face = "italic", size = 10),
          axis.text.x = element_text(size = 12),
          legend.position = "none", plot.title.position = "panel",
          title = element_text(size = 17),
          strip.text.y = element_text(size = 12, face = "bold"),
          strip.placement = "outside",
          panel.grid.major.y = element_blank(),
          ...)
}

# DATA IMPORT ----
sheets <- loadWorkbook('Data2024.xlsx') |>
  names() |>
  set_names()

data <- map(sheets, \(x) readWorkbook('Data2024.xlsx', sheet = x))
```

```

# "MOST FREQUENT SPECIES" plot ----

get_frequency <- function(x, cutoff = 0){
  x |>
    pivot_longer(!plotID) |>
    filter_out(value == 0) |>
    count(name) |>
    arrange(desc(n)) |>
    filter(n > cutoff) |>
    mutate(name = str_replace(name, "(?<=\\w)\\. (?=\\w)", " "))
}

## Plants data
# Tree layer

tree_freq <- data$GS_2024_trees |>
  get_frequency(cutoff = 1)

# Shrubs

shrub_freq <- data$GS_2024_shrubs |>
  get_frequency(cutoff = 1)

# Herbs

herb_freq <- data$GS_2024_herbs |>
  get_frequency(cutoff = 16)

## Lichens data

lichen_freq <- data$GS_2024_lichens |>
  get_frequency(cutoff = 14)

## Fungi data

fungi_freq <- data$GS_2024_fungi |>
  get_frequency(cutoff = 5)

## Dataset binding

freq_all <- bind_rows(tree_freq, shrub_freq, herb_freq, lichen_freq, fungi_freq, .id = "group") |>
  mutate(group = recode_values(group,
    "1" ~ "Tree\\nlayer",
    "2" ~ "Shrub\\nlayer",
    "3" ~ "Herb\\nlayer",
    "4" ~ "Lichens",
    "5" ~ "Fungi")) |>
  mutate(group = as_factor(group))

# "Most frequent species" Graph

gg_spp_freq <- freq_all |>
  ggplot(aes(x = fct_reorder(name, n), y = n, fill = group)) +
  geom_bar(stat = 'identity', width = 0.75) +

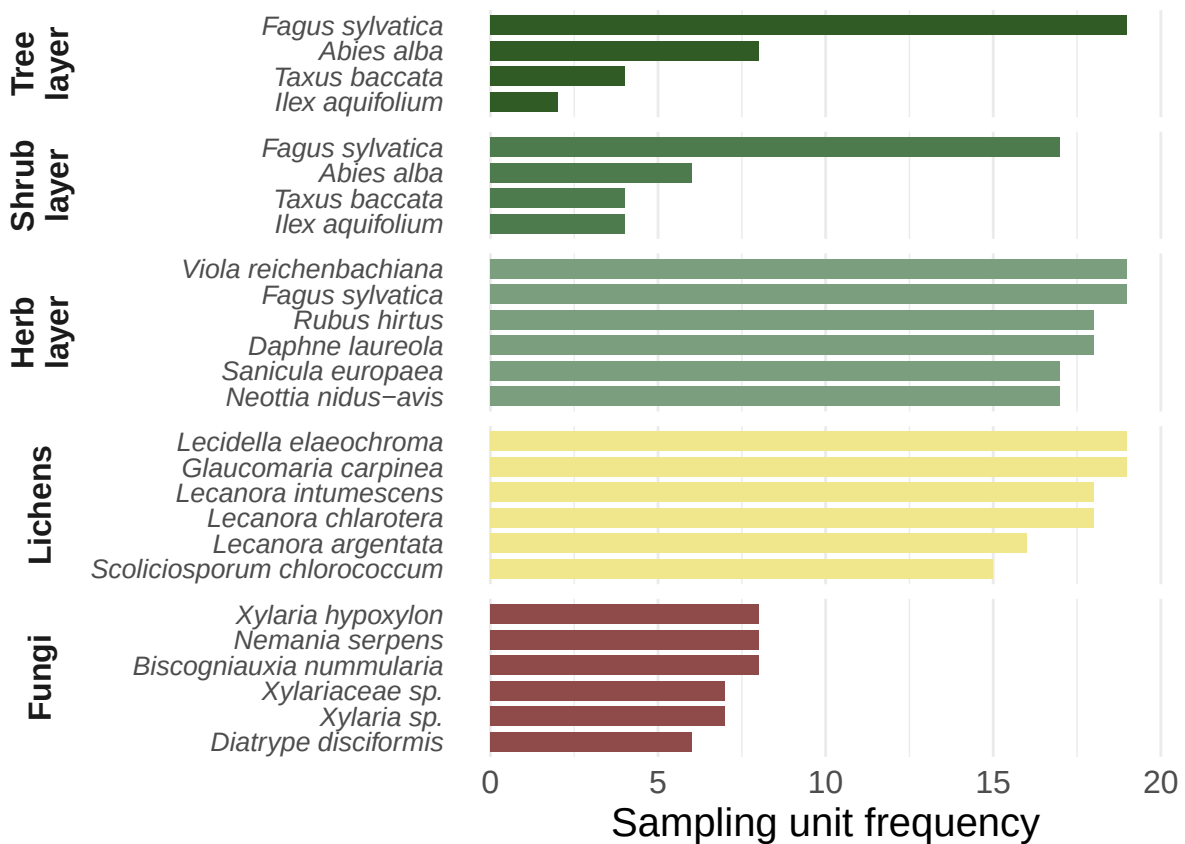
```

```

scale_fill_manual(values = c(
  "Tree\nlayer" = "#315b25",
  "Shrub\nlayer" = "#4d7b4d",
  "Herb\nlayer" = "#7B9E7E",
  "Lichens" = "#F0E68C",
  "Fungi" = "#8F4A4A")) +
coord_flip() +
labs(y = "Sampling unit frequency", x = "") +
ylim(c(0, 20)) +
facet_grid(group ~ ., scales = "free", space = "free",
  switch = "both") +
my_theme()

gg_spp_freq

```



```

# Save graph - EPS format
ggsave("Fig3.eps",
  plot = gg_spp_freq,
  width = 119,
  height = 73,
  scale = 1.64,
  units = "mm",
  bg = "white")

# Save graph - PNG format

```

```

ggsave(
  filename = "most_frequent_species.png",
  plot = gg_spp_freq,
  width = 119,
  height = 73,
  scale = 1.64,
  units = "mm",
  bg = "white")

# "FOREST STRUCTURE" plot ----
forest_str <- data$GS_2024_structure |>
  separate_wider_position(plotID, c(site = 2, plot = 1)) %>%
  mutate(site = recode_values(site,
                              "PR" ~ "Prati di Tivo",
                              "IN" ~ "Incodaro",
                              "VE" ~ "Venaquaro"))

# Number of records per each tree species in each study site
forest_str |>
  count(site, treesp) |>
  pivot_wider(names_from = site, values_from = n) |>
  arrange(treesp)

```

```

## # A tibble: 5 x 4
##   treesp      Incodaro 'Prati di Tivo' Venaquaro
##   <chr>      <int>      <int>      <int>
## 1 Abies alba      87          NA          NA
## 2 Acer pseudoplatanus    1          NA          NA
## 3 Fagus sylvatica    382        189        314
## 4 Ilex aquifolium      NA          NA          2
## 5 Taxus baccata      NA          8          1

```

```

# Creation of a common legend among study site
species_str <- forest_str |>
  distinct(treesp) |>
  arrange(treesp) |>
  pull()

species_str

```

```

## [1] "Abies alba"      "Acer pseudoplatanus" "Fagus sylvatica"
## [4] "Ilex aquifolium" "Taxus baccata"

```

```

col_leg <- scale_fill_manual(
  name = "Species",
  limits = species_str,
  drop = FALSE,
  values = c(
    "Abies alba" = "#0072B2",
    "Acer pseudoplatanus" = "#E69F00",
    "Fagus sylvatica" = "#56B4E9",
    "Ilex aquifolium" = "#000000",

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```

    "Taxus baccata" = "#D55E00"
  )
)

# Ranges, means, medians, quantiles and histograms of DBH in...

my_summary <- function(x, var){
  x |>
    summarise(
      min    = min({{var}}, na.rm = T),
      max    = max({{var}}, na.rm = T),
      mean   = mean({{var}}, na.rm = T),
      median = median({{var}}, na.rm = T),
      q25    = quantile({{var}}, probs = 0.25, na.rm = T),
      q75    = quantile({{var}}, probs = 0.75, na.rm = T)
    )
}

hist_str <- function(x, where, what, xlim, ylim, binwidth, xlab){
  x |>
    filter(site == where) |>
    ggplot(aes({{what}}, fill = treesp)) +
    geom_histogram(color = "#636363", binwidth = binwidth, linewidth = 0.2,
                  show.legend = T) +
    col_leg +
    labs(x = xlab, y = "Frequency", fill = "Species", title = where) +
    scale_x_continuous(limits = xlim, oob = scales::squish) +
    ylim(ylim) +
    theme_minimal() +
    theme(plot.title = element_text(hjust = 0.4),
          legend.text = element_text(face = "italic"),
          panel.grid.major.x = element_blank(),
          panel.grid.minor.x = element_blank())
}

# ...entire dataset

forest_str |>
  my_summary(treedb)

```

```

## # A tibble: 1 x 6
##   min    max  mean median  q25   q75
##   <dbl> <dbl> <dbl>  <dbl> <dbl> <dbl>
## 1    2.5   112  25.2    20    13  34.2

```

```

xlim_db <- range(forest_str$treedb)
ylim_db <- c(0, 100)

# ...by site

forest_str |>
  group_by(site) |>
  my_summary(treedb)

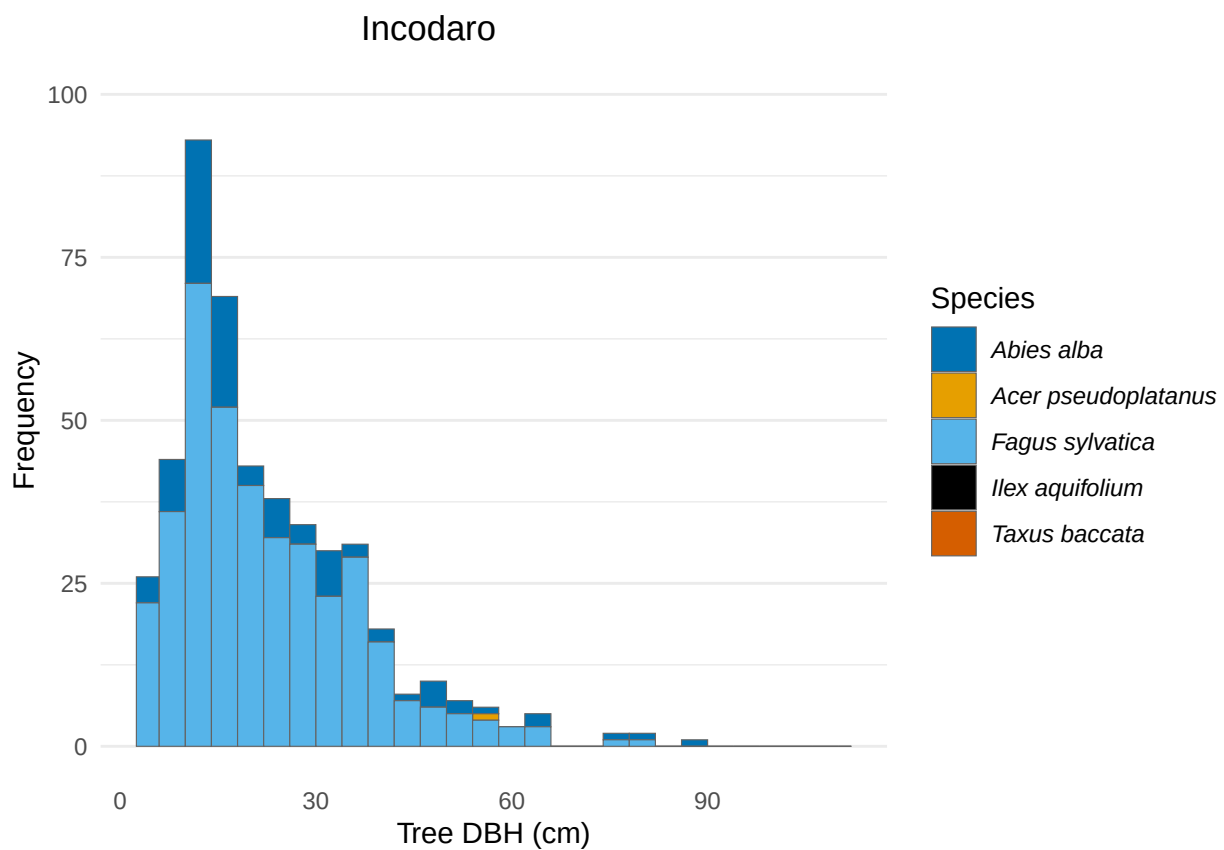
```

```
## # A tibble: 3 x 7
##   site      min    max  mean median   q25   q75
##   <chr>    <dbl> <dbl> <dbl>  <dbl> <dbl> <dbl>
## 1 Incodaro      3     88  23.1    19    12    31
## 2 Prati di Tivo  2.5   112  30.3    23    13    48
## 3 Venaquaro     3     73  25.3    22    14    34
```

```
# ... Incodaro study site
```

```
hist_IN <- forest_str |>
  hist_str("Incodaro", treedb, xlim_db, ylim_db, 4, "Tree DBH (cm)")
```

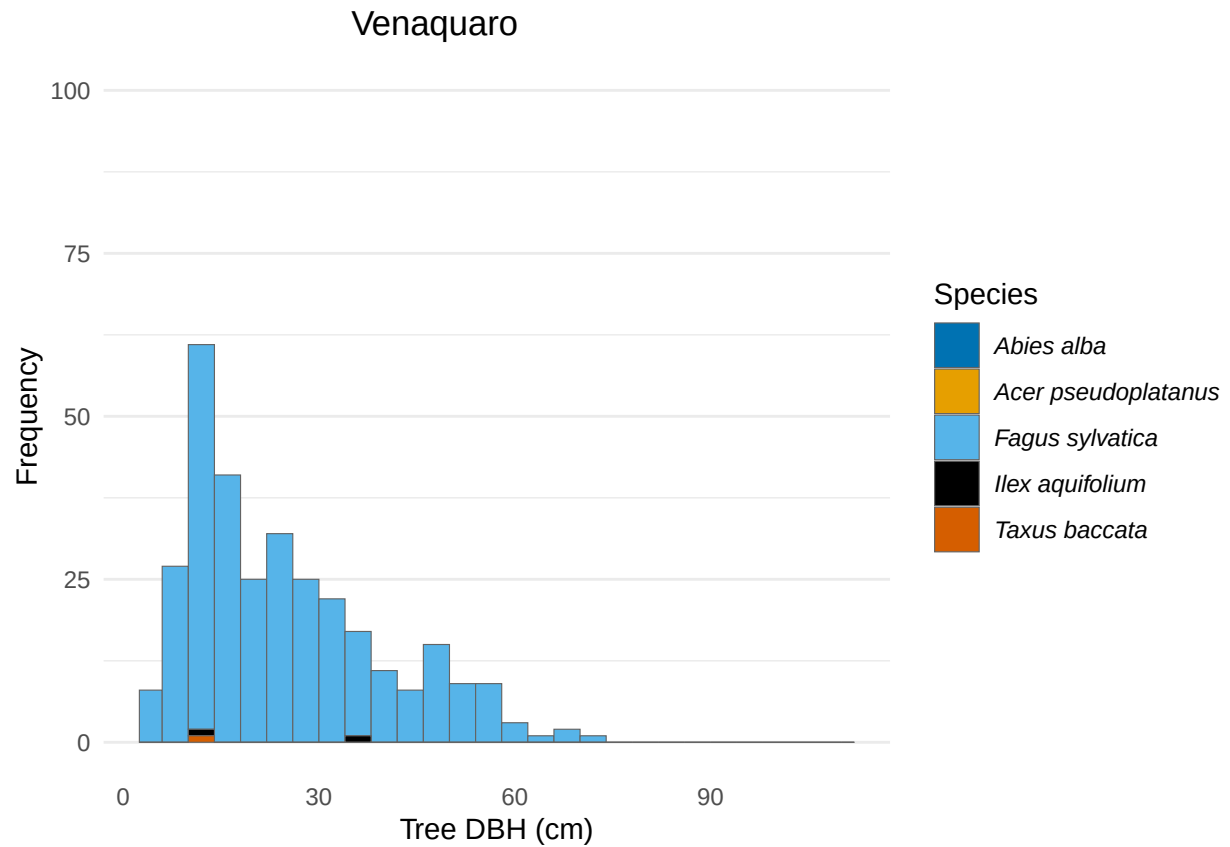
```
hist_IN
```



```
# ... Venaquaro study site
```

```
hist_VE <- forest_str |>
  hist_str("Venaquaro", treedb, xlim_db, ylim_db, 4, "Tree DBH (cm)")
```

```
hist_VE
```

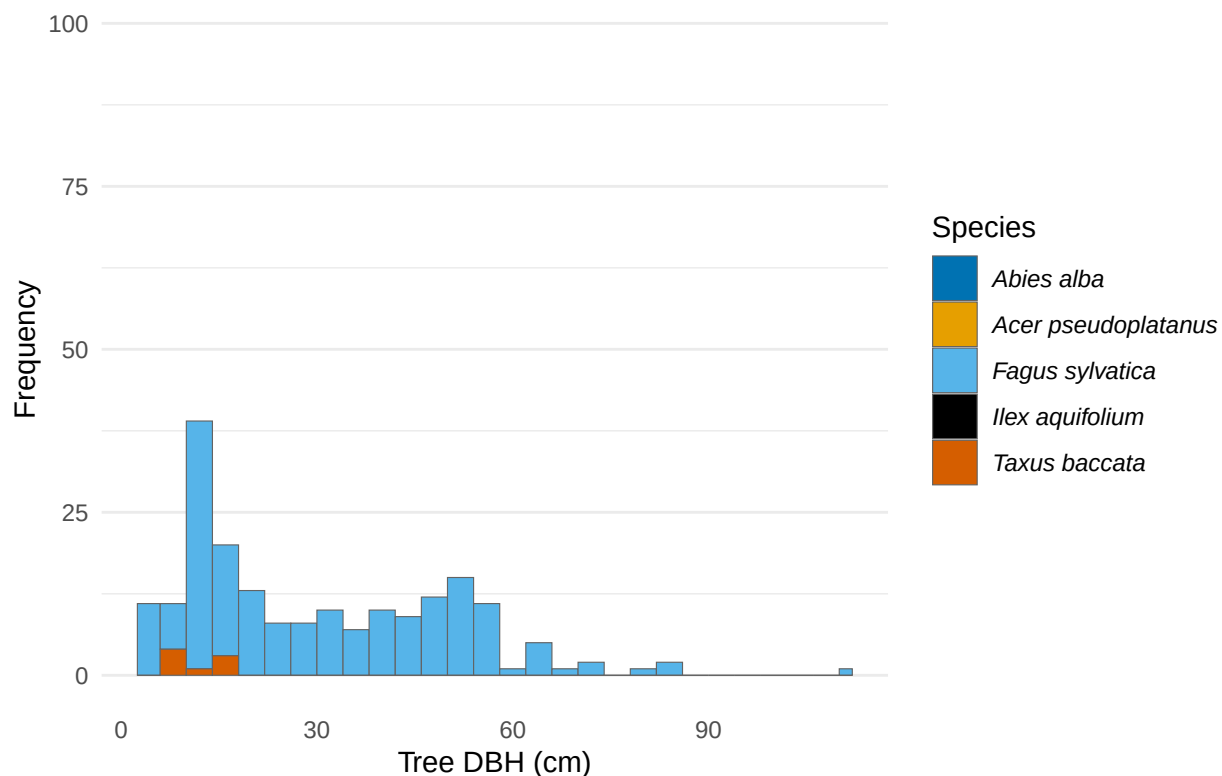


```
# ... Prati di Tivo study site
```

```
hist_PR <- forest_str |>
  hist_str("Prati di Tivo", treedb, xlim_db, ylim_db, 4, "Tree DBH (cm)")
```

```
hist_PR
```

Prati di Tivo



```
# Ranges, means, medians, quantiles and histograms of height measured in...
# ... entire dataset
forest_str |>
  my_summary(treeht)
```

```
## # A tibble: 1 x 6
##   min   max mean median  q25  q75
##   <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1     2   42 19.4  20.2   12   27
```

```
xlim_ht <- range(forest_str$treeht, na.rm = T)
ylim_ht <- c(0,13)
```

```
# ... by site
```

```
forest_str |>
  group_by(site) |>
  my_summary(treeht)
```

```
## # A tibble: 3 x 7
##   site      min   max mean median  q25  q75
##   <chr>    <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 Incodaro     2   42  19.4  20.8    8   28
## 2 Prati di Tivo 2.5  38  20.0  21.8  13.4  27
## 3 Venaquaro   2.5 35.1  19.0  19.7  13.5 24.2
```



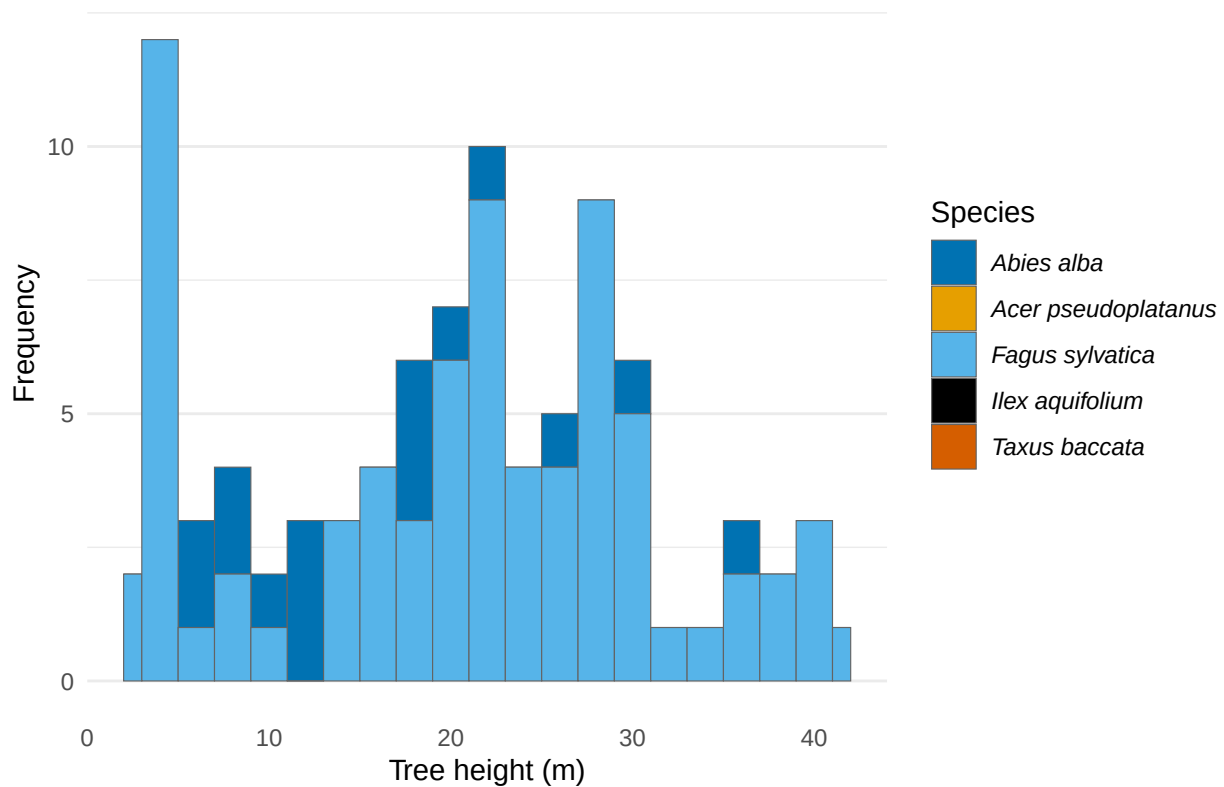
```
# ... Incodaro study site
```

```
hist_ht_IN <- forest_str |>
  hist_str("Incodaro", treeht, xlim_ht, ylim_ht, 2, "Tree height (m)") +
  ggtitle(NULL)
```

```
hist_ht_IN
```

```
## Warning: Removed 374 rows containing non-finite outside the scale range
## ('stat_bin()').
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_bar()').
```

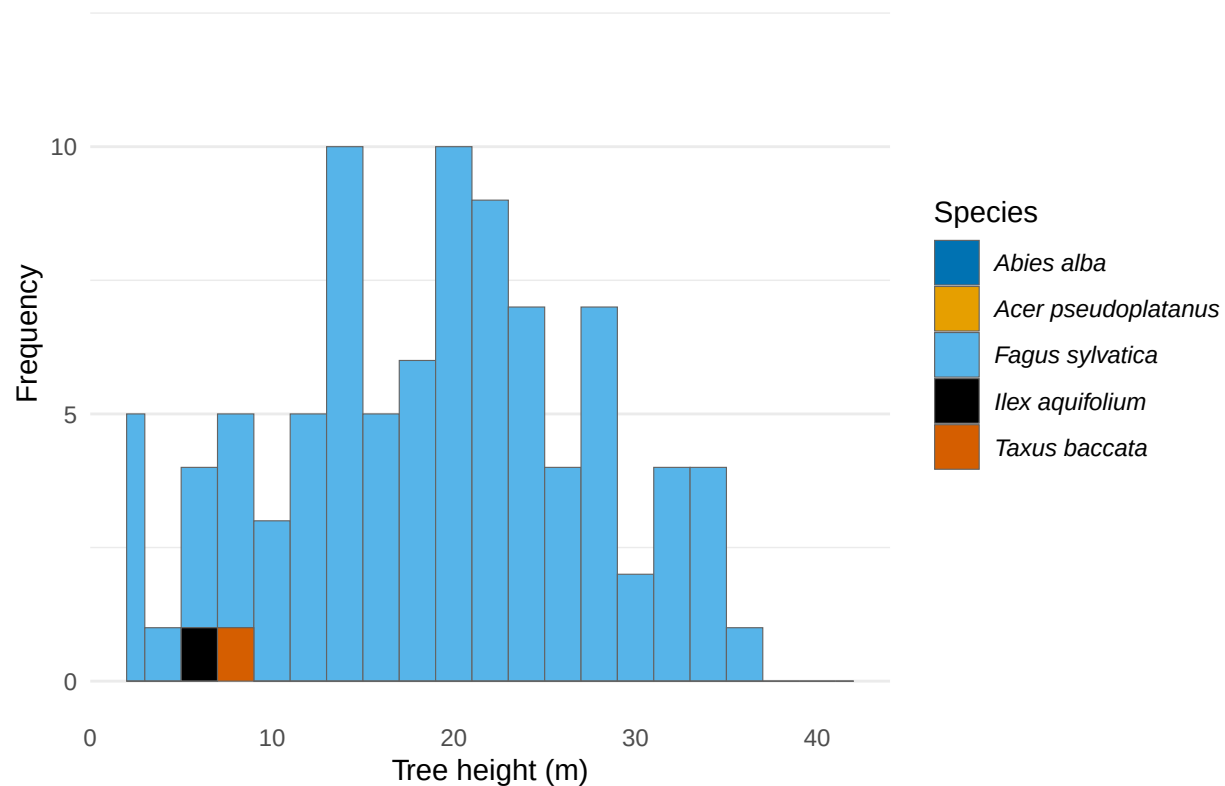


```
# ... Venaquaro study site
```

```
hist_ht_VE <- forest_str |>
  hist_str("Venaquaro", treeht, xlim_ht, ylim_ht, 2, "Tree height (m)") +
  ggtitle(NULL)
```

```
hist_ht_VE
```

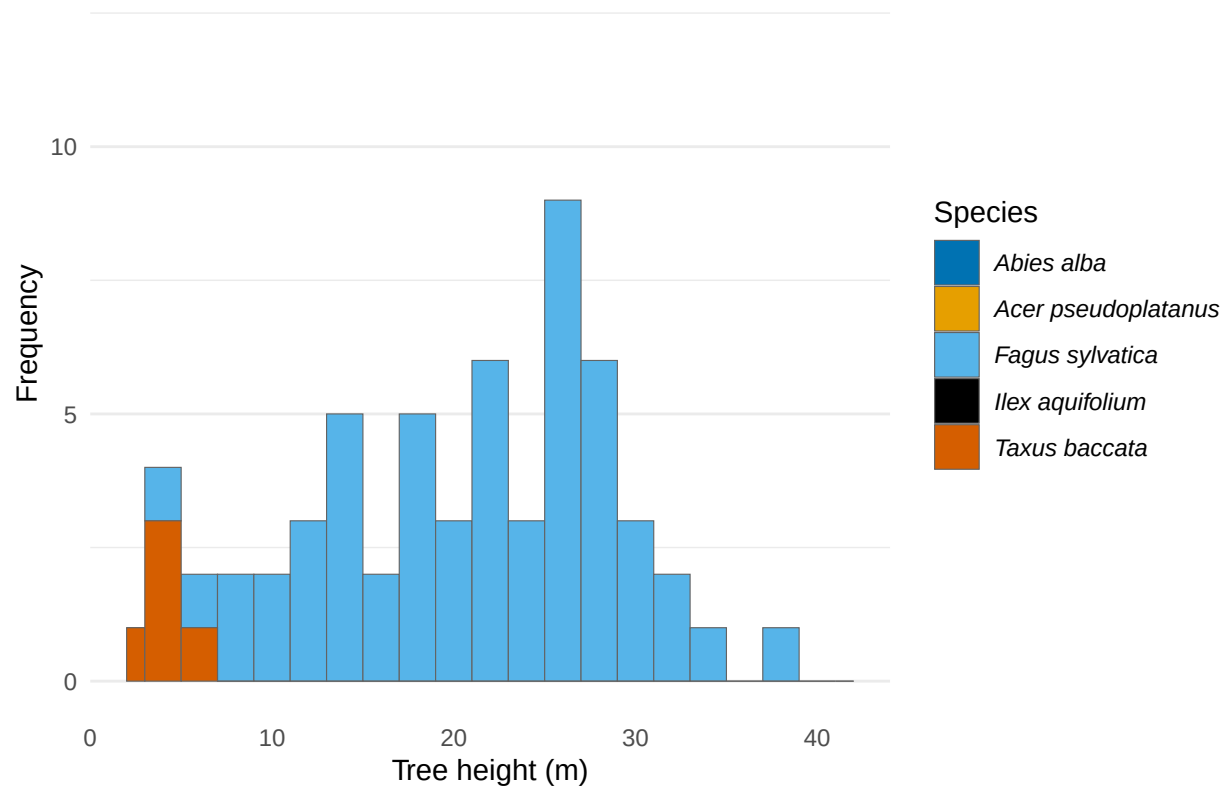
```
## Warning: Removed 225 rows containing non-finite outside the scale range
## ('stat_bin()').
```



```
# ... Prati di Tivo study site
hist_ht_PR <- forest_str |>
  hist_str("Prati di Tivo", treeht, xlim_ht, ylim_ht, 2, "Tree height (m)") +
  ggtitle(NULL)

hist_ht_PR
```

```
## Warning: Removed 137 rows containing non-finite outside the scale range
## ('stat_bin()').
```



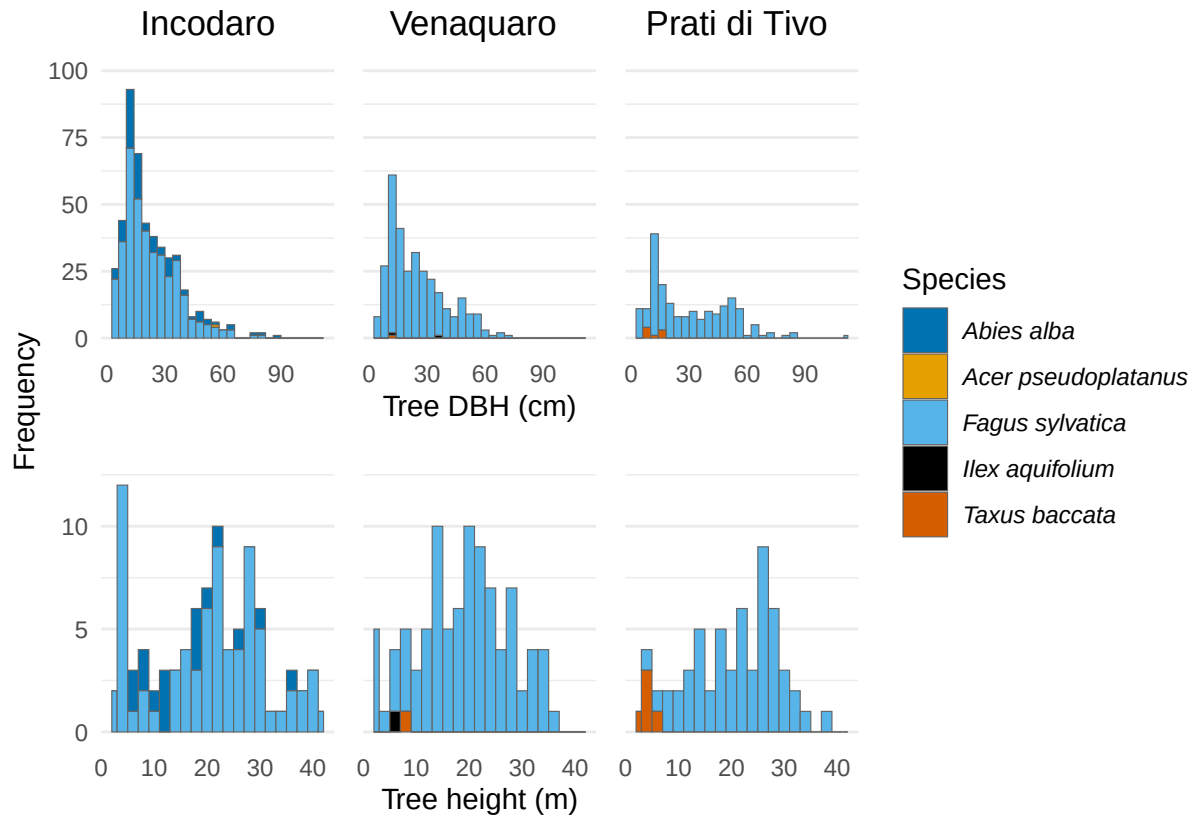
```
# Combined DBH and heights histograms of the three sites
combi_plots <- wrap_plots(hist_IN, hist_VE, hist_PR,
                           hist_ht_IN, hist_ht_VE, hist_ht_PR,
                           ncol = 3, nrow = 2,
                           guides = "collect", axes = "collect")
```

```
combi_plots
```

```
## Warning: Removed 374 rows containing non-finite outside the scale range ('stat_bin()').
## Removed 1 row containing missing values or values outside the scale range
## ('geom_bar()').
```

```
## Warning: Removed 225 rows containing non-finite outside the scale range
## ('stat_bin()').
```

```
## Warning: Removed 137 rows containing non-finite outside the scale range
## ('stat_bin()').
```



```
# Save graph - EPS format
```

```
ggsave("Fig4.eps",
  combi_plots,
  width = 119,
  height = 73,
  scale = 1.7,
  units = "mm",
  bg = "white")
```

```
## Warning: Removed 374 rows containing non-finite outside the scale range
## ('stat_bin()').
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_bar()').
```

```
## Warning: Removed 225 rows containing non-finite outside the scale range
## ('stat_bin()').
```

```
## Warning: Removed 137 rows containing non-finite outside the scale range
## ('stat_bin()').
```

```
# Save graph - PNG format
```

```
ggsave(
  filename = "forest_structure.png",
```

```

combi_plots,
width = 119,
height = 73,
scale = 1.7,
units = "mm",
bg = "white")

## Warning: Removed 374 rows containing non-finite outside the scale range
## ('stat_bin()').

## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_bar()').

## Warning: Removed 225 rows containing non-finite outside the scale range
## ('stat_bin()').

## Warning: Removed 137 rows containing non-finite outside the scale range
## ('stat_bin()').

# DEADWOOD table ----

# entire dataset
data$GS_2024_deadwood |>
  pivot_longer(diam1:height) |>
  group_by(name) |>
  my_summary(value)

## # A tibble: 4 x 7
##   name      min    max  mean median   q25   q75
##   <chr> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 diam1      8    110  22.5    17    12    28
## 2 diam2      8     99  31.7    29    19    40
## 3 height     4    120  23.2    16    12    25
## 4 length    10   1800 199.    110    91   190

#by site
data$GS_2024_deadwood |>
  pivot_longer(diam1:height) |>
  mutate(siteID = str_sub(plotID, 1, 2)) |>
  select(!c(plotID, lynID)) |>
  group_by(name, siteID) |>
  my_summary(value)

## 'summarise()' has regrouped the output.
## i Summaries were computed grouped by name and siteID.
## i Output is grouped by name.
## i Use 'summarise(groups = "drop_last")' to silence this message.
## i Use 'summarise(.by = c(name, siteID))' for per-operation grouping
## ('?dplyr::dplyr_by') instead.

```

```
## # A tibble: 12 x 8
## # Groups:   name [4]
##   name  siteID  min  max  mean median  q25  q75
##   <chr> <chr> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 diam1 IN      8   110  19.0   15    11    22
## 2 diam1 PR      8   110  30.6   20    12   44.8
## 3 diam1 VE      8    61  26.3   22    13   36.2
## 4 diam2 IN     11.5   64  28.9   30   16.2   37
## 5 diam2 PR     11    70  40.2   40   26.5   52.2
## 6 diam2 VE      8    99  29.6   26   17.2   39.8
## 7 height IN      4    73  18.3   15    11    20
## 8 height PR      5   120  30.2   16   11.8   35
## 9 height VE      5   110  23.3   19.5  12.2   26
## 10 length IN     10  1800  210.   110   96.2  190
## 11 length PR     16   640  137.    95    66   150
## 12 length VE     13   700  198   137    70   295
```